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# DAY-4

## TASK 5: DIFFERENCE BETWEEN OPENPOSE AND MEDIAPIPE

OpenPose is a real-time human pose estimation framework used to detect and track human body keypoints such as the head, shoulders, elbows, wrists, hips, knees, and ankles from images or video streams.

It may appear same as MediaPipe in many ways but there a lot of differences between the both.

## DIFFERENCES ARE:

### 1. Accuracy

OpenPose generally provides higher pose detection accuracy and more detailed landmark tracking. MediaPipe also gives good accuracy but is mainly optimized for real-time performance.

### 2. Speed and Performance

MediaPipe is faster and lightweight, making it suitable for real-time applications on normal systems. OpenPose is comparatively slower because it performs more complex computations.

### 3. Hardware Requirements

OpenPose usually requires a GPU for smooth performance and better processing speed. MediaPipe can work efficiently on CPUs and mobile devices without needing high-end hardware.

### 4. Multi-Person Detection

OpenPose performs better in detecting and tracking multiple people simultaneously. MediaPipe is mainly designed for single-person tracking and has limited multi-person support.

**5. Ease of Use**

MediaPipe is beginner-friendly and easier to install and integrate with Python and OpenCV projects. OpenPose setup is more complex and may require additional dependencies and configuration.

**6. Real-Time Applications**

MediaPipe is widely used in applications like virtual try-on, fitness tracking, and mobile AI because of its speed. OpenPose is more commonly used in research projects and advanced motion analysis systems.